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STEPANBLEND® ASF/MB

Ready, Easy, Sulfate free

Agenda

1. **Empowering formulators: One blend, endless possibilities**
   STEPANBLEND® ASF/MB: one ready-to-dilute, sulfate-free blend
2. **Foaming & viscosity building**
   Foam performance, « no salt added » viscosity and starter formulas
3. **Color care, combability & mildness**
   Fade resistance, combability and skin-mildness substantiation
4. **Key Take Aways**
   Summary of the proven wins

Empowering formulators: one blend, endless formulas

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STEPANBLEND® ASF/MB — Technical data

RSPO mass balance-certified, 39% solids blend for mild cleansing formulations

WHAT IT IS?

A ready-to-dilute surfactant blend, RSPO mass balance-certified, designed to build mild, high-performing rinse-off formulations.

FREE FROM

- Sulfated surfactants
- Diethanolamine (DEA)
- Ethylene Oxide (EO)
- Preservatives

39% Solids	**11.2** pH (10% aqueous)	**17,000 cps** Viscosity at 20°C
Liquid Form at 20°C	**1.05** Density at 20°C	**< 5°C** Freeze point

HANDLING

Cold-processable, no heating required · dilute directly in water at 14% to 39% depending on the format.

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Multiple reasons to go « Naturally Simple » with STEPANBLEND® ASF/MB

One ready-to-dilute blend — faster to formulate, easier to make, proven on hair and skin.

Shorter Formulation Development

Finished formula fast, ISO 16128 naturality up to 97% with proven performance

Less Chemical Inventory

Four surfactants, one drum — broaden your portfolio with one SKU only

Shorter Production Time

Cold-processable, fewer additions, real energy saving

No Salt Added

Viscosity adjusted simply for higher viscosities formulations

1 Blend, endless possibilites

Hand wash, shower gel, shampoo 5 to 14% actives

Gentle and claim-ready

Color fade resistance, improved combability, low Zein score and Non-irritating on human skin (M.C.I.I. below 0.25 on human patch test.

Savings all the way down to market launch

Lab — fewer trials, faster to a finished formula

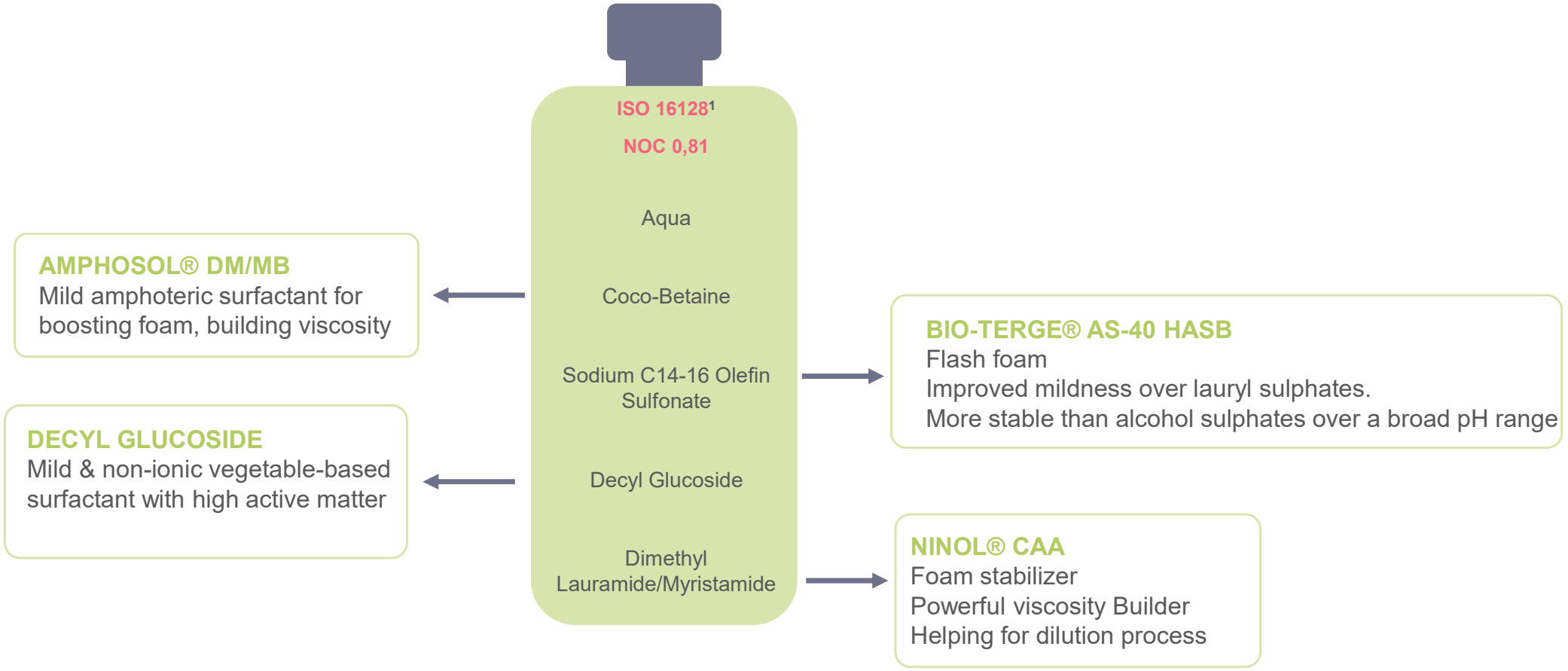
Sourcing — fewer raw materials to buy, qualify and store

Production — cold process, no heating, no salt addition

Launch — one blend serving a whole portfolio, faster to market

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A ready-to-dilute, sulfate-free blend of chosen surfactants



[1]Formulation Chassis Naturality = % of the ingredients multiplied by Origin Index/Content (NOI/NOC) Calculation according to ISO 16128
Disclaimer: ISO 16128: Stepan interprets the Natural Origin Index (NOI) under ISO 16128 to be calculated based on the biorenewable carbon approach allowed by the norm.
We take water into account (NOI=1), additives and known impurities. Stepan assumes all raw materials and processes used to manufacture our products to be accepted by the norm.

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One Simple Blend, Endless formulas

1 ready-to-dilute blend

Cold-processable
High-foaming

For hand/face washes, shower gels and shampoos.

Endless possibilities

These three formulas are just the starting point.

Adapt actives, viscosity, pH and fragrance. Add conditioning agents and claim ingredients.

Build formulation variants to suit your portfolio.

Mild, Sulfate-Free Hand/Face Wash



5%
actives

14% blend in water

ISO 16128
NOC 0,97

Mild, Sulfate-Free Shower Gel



10%
actives

27% blend in water

ISO 16128
NOC 0,95

Mild, Sulfate-Free Shampoo



14%
actives

39% blend in water

ISO 16128
NOC 0,92

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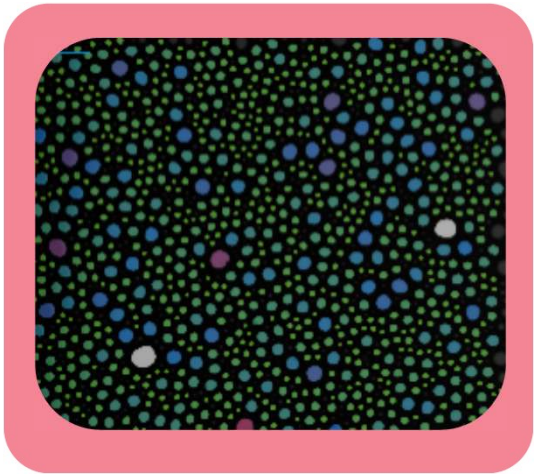
Formulating for
Foaming & Viscosity Building

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Maximizing Foam with STEPANBLEND® ASF/MB

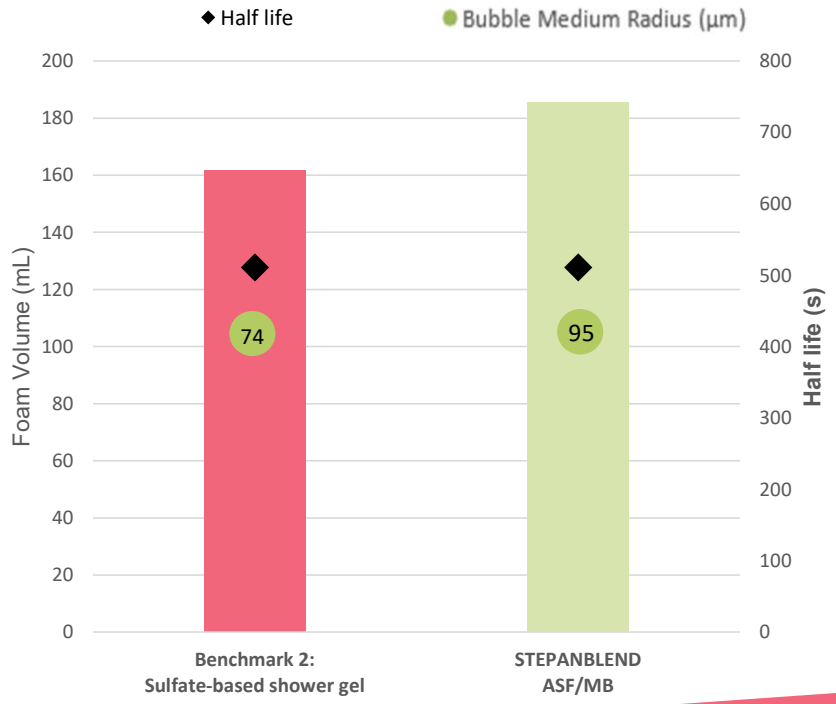
STEPANBLEND® ASF/MB provides excellent foam volume and similar foam quality.

SLES & CAPB
Sulfate

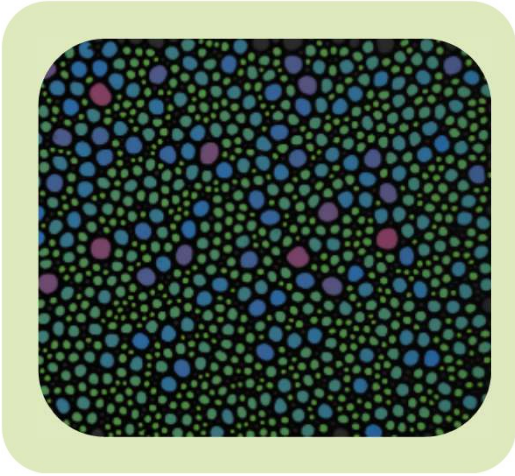


Foaming Properties

Conditions: 10% actives – Foam Scan analyzer



STEPANBLEND® ASF/MB
Sulfate-free



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Maximizing Viscosity with STEPANBLEND® ASF/MB

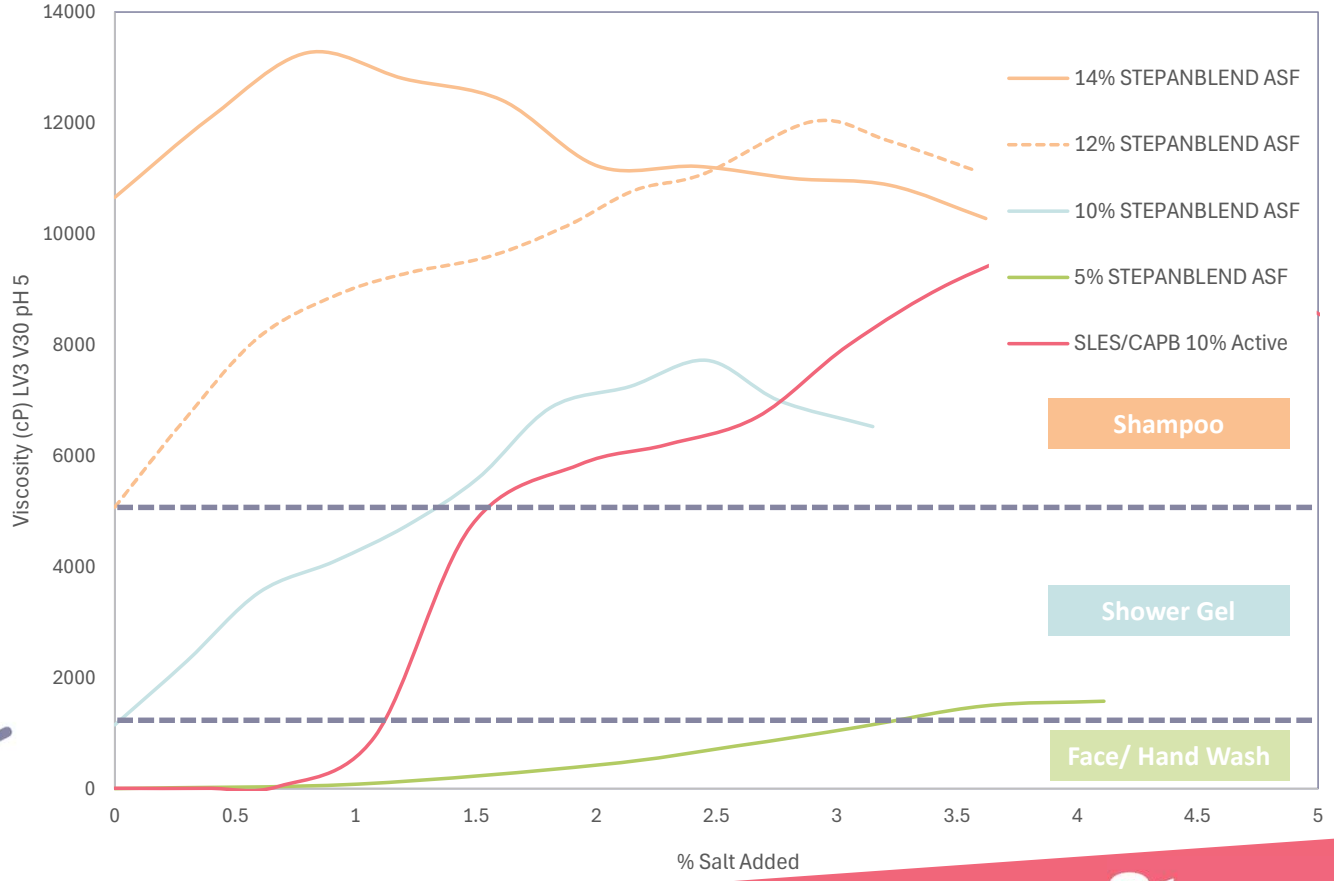


STEPANBLEND® ASF/MB

Enables « no salt added » formulations

Easier viscosity adjustment

Recommended active usage of STEPANBLEND® ASF/MB from 5 to 14% for typical viscosity level for rinse off cosmetics



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Formulation Inspirations



Hand Wash, Made Easy

FORMULATION SERIES

Phase	Ingredients	(w/w) %
A	Aqua	78,1
	STEPANBLEND®® ASF/MB	14,1
B	Glycerin	5,0
	Sodium Benzoate (and) Potassium Sorbate	1,2
	Citric Acid	0,9
	Fragrance	0,2
C	Sodium Chloride	0,5
D	Dye	qs

Process

- In a vessel, introduce STEPANBLEND® ASF/MB in deionized water
- Cold water can be used but process will be faster with warm water
- Add the other products in the given order
- Adjust pH between 5 and 5,5
- Adjust Sodium Chloride amount to get desired viscosity

pH, as is	5,3
Viscosity @ 25°C (Brookfield LV1/V12)	1690 cP
Appearance	Transparent thin gel

"Super Easy-to-Make" Mild, Sulfate-Free Hand Wash

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Shower Gel, Made Easy

FORMULATION SERIES

Phase	Ingredients	(w/w)%	Process
A	Aqua	69,5	• In a vessel, introduce STEPANBLEND® ASF/MB in deionized water • Cold water can be used but process will be faster with warm water • Add the other product in the given order • Adjust pH between 5 and 5,5 • Adjust STEPAN-MILD GCC-FL/MB amount to get desired viscosity, and condition feel enhancer
	STEPANBLEND®® ASF/MB	28,2	
B	Sodium Benzoate (and) Potassium Sorbate	0,6	
C	Citric acid	1,0	
D	Fragrance	0,2	
E	**STEPAN-MILD®® GCC-FL/MB** Glyceryl caprylate/caprate (and) Glycerin (and) Propanediol	0,5	

pH, as is	5,4
Viscosity @ 25°C (Brookfield LV2/V30)	4020 cP
Appearance	Transparent thin gel

"Super Easy-to-Make" Mild, Sulfate-Free Shower Gel

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Shampoo, Made Easy

FORMULATION SERIES

Phase	Ingredients	(w/w) %	Process
A	Aqua	58,5	• In a vessel, introduce STEPANBLEND® ASF/MB in deionized water • Cold water can be used but process will be faster with warm water • Add the other product in the given order • Adjust pH between 5 and 5,5 • Adjust STEPAN-MILD L3-G/MB amount to get desired viscosity, and condition feel enhancer
	STEPANBLEND®® ASF/MB	39,4	
B	Sodium Benzoate (and) Potassium Sorbate	0,6	
	Citric Acid	1,0	
	Fragrance	0,2	
	STEPAN-MILD® L3-G/MB Lauryl Lactyl Lactate	0,3	
	Dye	Qs	

pH, as is	**5,4**
Viscosity @ 25°C (Brookfield LV2/V12)	**13200 cP**
Appearance	**Transparent thin gel**

"Super Easy-to-Make" Mild, Sulfate-Free Shampoo

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Pop-Up Video

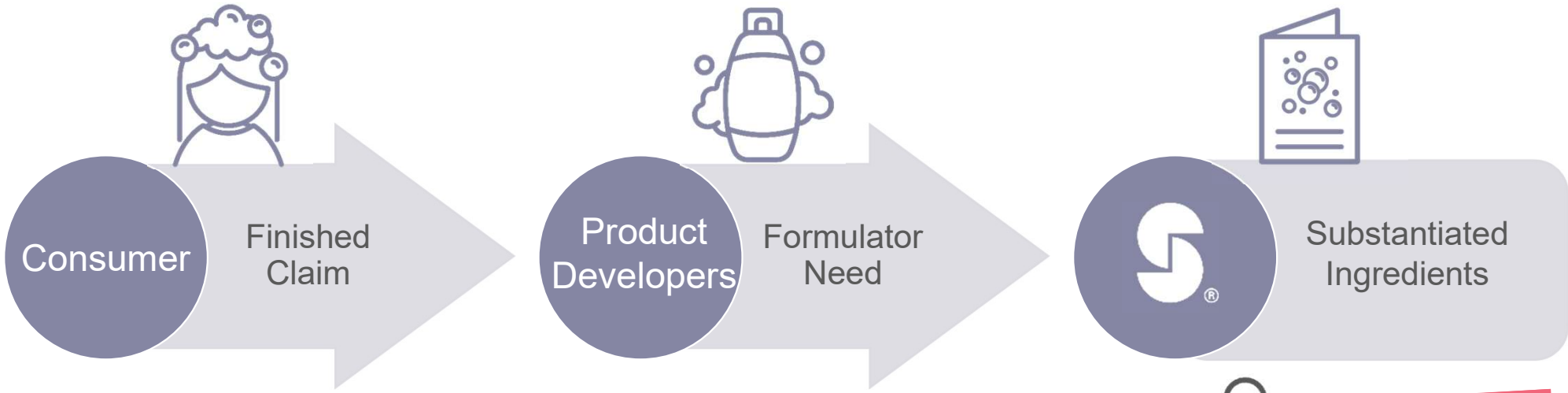
“Super Easy To Make”
Mild Sulfate-Free Shampoo
Formulation No. 1541



Formulating for
Color Care, Combability & Mildness

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Formulating for top shampoo claims



Color Care
Smooth Hair
Mildness

Fade Resistance
Combability
Gentle





Combing Force Assessment - DIASTRON

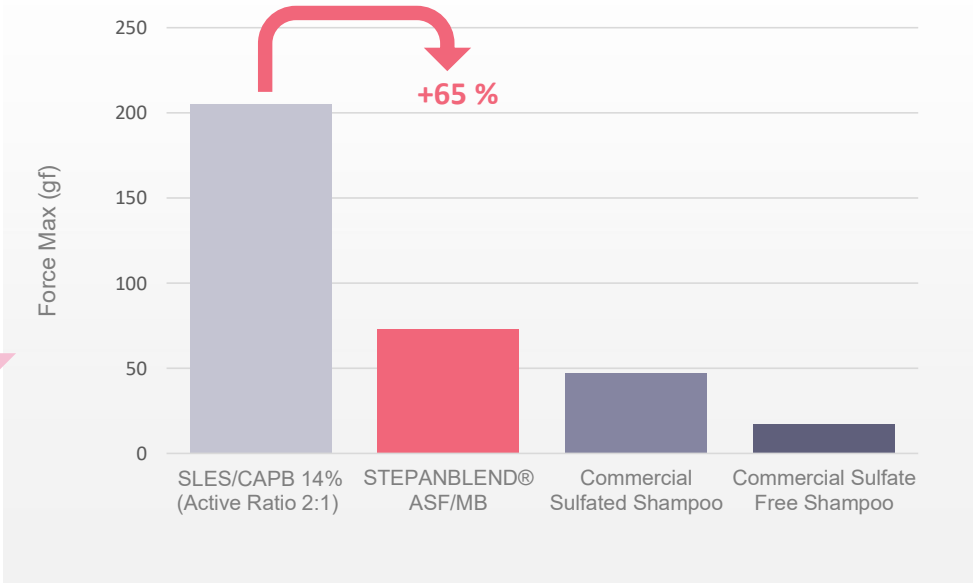
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MC1

Support gentle detangling with mild care

Combability

STEPANBLEND® ASF/MB improved dry combability compared to SLES:CAPB for better manageability.



Combing Force was measured at the end of the washing cycles (15 washes). Combing Force is quantified as the maximum amount of force necessary to comb the hair tress.

Method: 6 hair tresses per prototype , 10 assessments per hair tress / 60 data points per prototype.

What's behind the benchmarks?

Commercial Sulfate-Free Shampoo

Guar + Polyquaternium

Commercial Sulfated Shampoo

Guar + Polyquaternium + Dimethicone

STEPANBLEND® ASF/MB

No conditioning polymer, no silicone

…yet STEPANBLEND® ASF/MB already performs close to these conditioned benchmarks — without them.

Add STEPAN MILD GCC-FL/MB

Frizz taming • shine • combability — a smart add to the formulator's toolbox

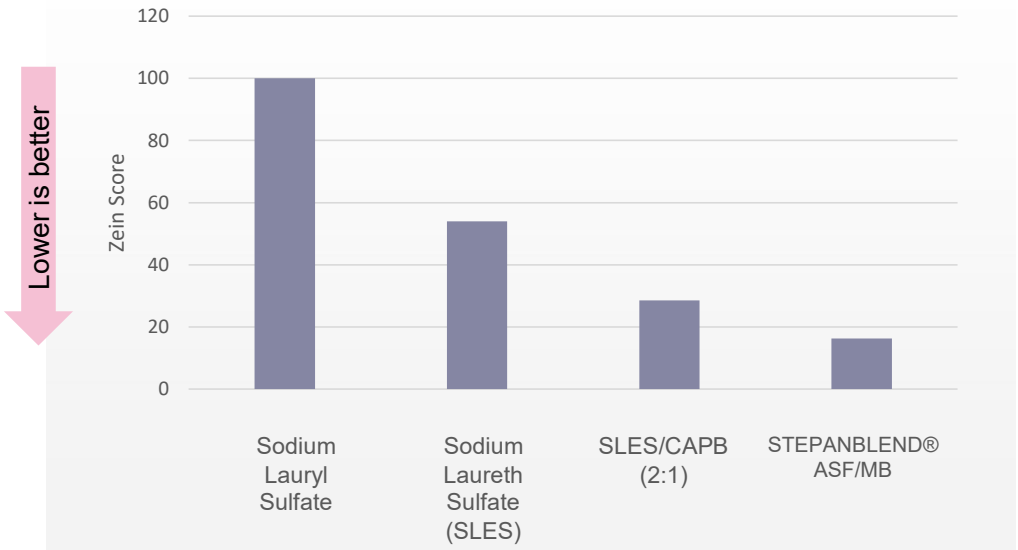
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Ultra Mild cleansing validated on human skin

Zein Score Mildness

STEPANBLEND® ASF/MB offers exceptional mildness for shampoos having a significantly lower Zein Score than traditional sulfate surfactants



Zein score test on 1% active surfactant solution. Scores are normalized to sodium lauryl sulfate, which is given a score of 100.

Mildness on Human Skin

10% actives
27% blend
in water
ISO 16128
NOC 0,95

48 hr Human Occlusive Patch test
Single application under patch-test on 21 Subjects

Human skin irritation test demonstrated that the prototype containing STEPANBLEND® ASF/MB at 10% actives[1] has a **M.C.I.I score below 0.25** as today's market leading baby shampoo, and ultra mild sulfate free blend benchmark classifying the prototype as **non irritating (NI)** following the scale of M.C.I.I[2]

Formulated with very mild surfactants:
coco betaine (AMPHOSOL DM/MB) and decyl glucoside

Deliberate inversion of the primary (BIO-TERGE AS-40 HASB) and secondary (AMPHOSOL DM/MB) surfactant ratio to reach a non-irritating level on skin

1. Further diluted at 5% in water according to intended use of wash-off products
2. M.C.I.I Mean Cumulative Irritation Index (M.C.I.I) scale
   M.C.I.I. < 0.25 Non irritating (NI); 0.25 ≤ M.C.I.I. < 0.50 Very slightly irritating (VSI); 0.5 ≤ M.C.I.I. < 1 Slightly irritating (SI); 1 ≤ M.C.I.I. < 2 Moderately irritating (MI); M.C.I.I. ≥ 2 Irritating (I)

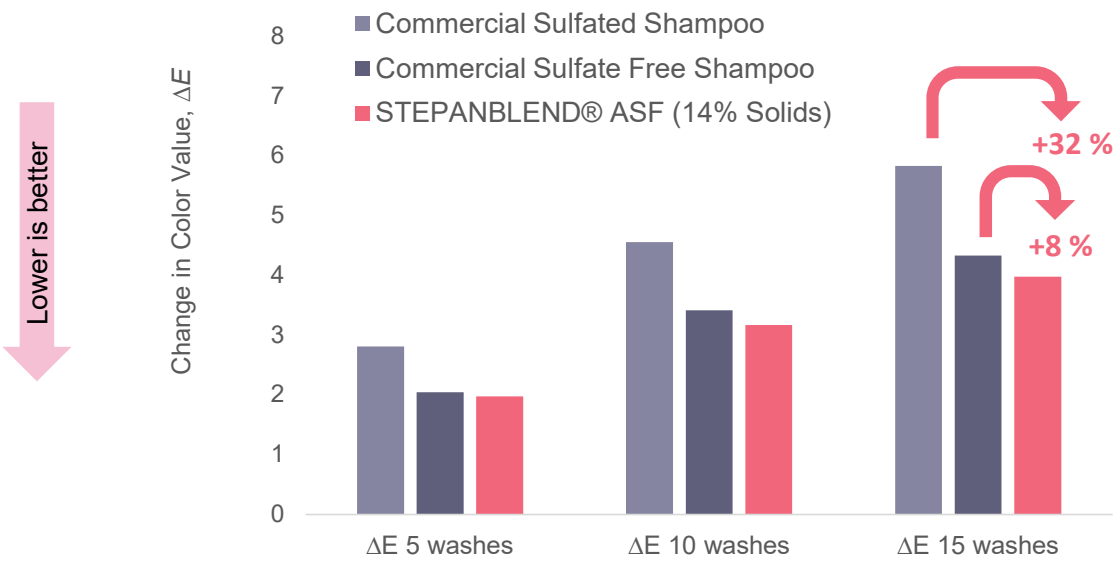
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Color Fading
Resistance

Protect colored hair with wash longevity

Fade Resistance

STEPANBLEND® ASF/MB delivers exceptional fade resistance, matching that of a leading commercial sulfate-free shampoo, with superior performance after multiple washes.





Hair tresses washing, Type 2A bleached with red shade, compared to untreated, 6 tresses per treatment.

Hair Tresses Type 2A bleached, moderately damaged hair. 0.1 g of shampoo applied per gram of hair tress weight.
1 min shampooing

Measurements made with Ultra Scan Vis Hunter Lab ColorimeterΔE=Delta E (ΔE) is a numerical measure of how different two colors appear, based on their distance in the CIELAB color space; a ΔE below 1 is typically imperceptible, while values above 3 indicate clearly visible differences. (unwashed).

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Key Take Aways

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One blend with proven wins

One blend = 4 surfactants

Ready-to-dilute, cold-processable — less inventory, faster to finished formula.

High foam & viscosity

Equal or better than SLES/CAPB, « no salt added » viscosity, 5–14% actives.

Mildness

Low Zein score and M.C.I.I. below 0.25 — milder than sulfated chassis.

Naturality

Sulfate-free chassis with ISO 16128 naturality up to 97%.

Hair care claims

Color fade resistance and improved dry combability for smooth, protected hair.

Simpler formulation, faster to market, claims you can substantiate

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